CSE445 Lecture 4.1

Recommender Systems: SVD-Based Collaborative Filtering

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Important Concepts

User Rating Matrix, Latent Factors, Rating Prediction, Similar Users, Similar Items

1. Why Recommender Systems?

A recommender system predicts what a user may like next. It compares available items, ranks them by relevance, and recommends the best matches.

Too many choices $\Rightarrow$ personalized ranking

Examples:

- Movies a user may view
- Online products a customer may buy
- Blog articles a reader may read

Key goal: Estimate unknown preferences for items the user has not rated or interacted with.

2. Two Traditional Approaches

Content-Based vs. Collaborative Filtering

Content-based recommendation recommends items by studying the features of the item itself.

Example: To recommend poems, the system may analyze their words, themes, and style. If a user likes romantic poems with simple language, the system suggests similar poems.

Collaborative filtering recommends items by studying the behavior of users.

Instead of analyzing the item text directly, it looks at who liked what, who rated what, or which users behaved similarly.

Main CF idea: Similar users like similar items, and similar items attract similar users.

3. Collaborative Filtering as a Utility Matrix

Assume there are m users and n items.

$$M \in \mathbb{R}^{m \times n}, \qquad M_{ij} = \text{rating or preference of user } i \text{ for item } j$$

	Movie 1	Movie 2	Movie 3	Movie 4
User A	5	4	?	2
User B	4	?	5	1
User C	?	2	4	5

The question mark is the recommendation problem: predict missing ratings from known ratings.

4. User-Based vs. Item-Based CF

User-based CF: compare the target user with other users.

$$\hat{r}_{u,i} \leftarrow \text{ratings from users similar to } u$$

Similarity can be measured using Pearson correlation or cosine similarity.

Item-based CF: compare the target item with items the user already rated.

$$\hat{r}_{u,i} \leftarrow \text{ratings of items similar to } i$$

Trade-off: User preferences may change over time; item relationships are often more stable, but hard to calculate.

5. SVD Breaks the Utility Matrix into Latent Factors

User ratings matrices are usually sparse: most users rate only a few items. SVD gives a low-rank approximation of the rating matrix.

$$M \approx U_r \Sigma_r V_r^T$$

$$\underbrace{M}_{\text{ratings}} \approx \underbrace{U_r}_{\text{user preferences}} \underbrace{\Sigma_r}_{\text{factor strength}} \underbrace{V_r^T}_{\text{item features}}$$

Part	Conceptual meaning
U_r	how strongly each user prefers each hidden factor
Σ_r	how important each hidden factor is overall
V_r^T	how strongly each item is related to each hidden factor

Hidden factors may behave like action, romance, comedy, or serious drama, even if we never named them manually.

6. Small Example: Preference $\times$ Strength $\times$ Item Features

Suppose SVD discovers two hidden factors: Action-like taste and Romance-like taste.

$$M \approx \underbrace{\begin{bmatrix} 0.9 & 0.2 \\ 0.8 & 0.3 \\ 0.1 & 0.9 \end{bmatrix}}_{U_r\text{: users} \times \text{factors}} \underbrace{\begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}}_{\Sigma_r\text{: factor strengths}} \underbrace{\begin{bmatrix} 0.9 & 0.8 & 0.7 & 0.1 \\ 0.1 & 0.2 & 0.3 & 0.9 \end{bmatrix}}_{V_r^T\text{: factors} \times \text{items}}$$

U_r says User A is mostly Action-oriented: $(0.9, 0.2)$. V_r^T says Movie 4 is mostly Romance-oriented: $(0.1, 0.9)$. Σ_r says the Action factor has stronger global influence than the Romance factor.

7. How SVD Predicts a Missing Rating

To predict User A's missing rating for Movie 3, use the user vector, the factor strengths, and the item vector.

$$\hat{M}_{A,3} = \underbrace{\begin{bmatrix} 0.9 & 0.2 \end{bmatrix}}_{\text{User A preferences}} \underbrace{\begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}}_{\text{factor strengths}} \underbrace{\begin{bmatrix} 0.7 \\ 0.3 \end{bmatrix}}_{\text{Movie 3 features}}$$

$$\hat{M}_{A,3} = (0.9)(5)(0.7) + (0.2)(3)(0.3) = 3.33$$

The predicted rating is high when the movie's hidden features match the user's hidden preferences. The value does not have to be exact; it is mainly used to rank candidate items.

8. Preference, Rating, and Similarity in One View

Concept	Meaning in SVD-based collaborative filtering
Preference	A user's position in the latent taste space; rows of U_r describe users
Rating	A predicted score from matching a user vector with an item vector through Σ_r
Item similarity	Items with similar columns in V_r^T have similar latent features
User similarity	Users with similar rows in U_r have similar latent preferences

So SVD changes the problem from comparing long, sparse rating rows to comparing short, dense latent vectors. This is why it helps with sparsity, prediction, and similarity search.

Important note: In real recommender systems, we either fill/normalize missing values first or learn the low-rank factors only from observed ratings.

9. Main Takeaways

Collaborative filtering learns from behavior.
SVD learns hidden structure.

1. A recommender system predicts future preference and ranks items.
2. Collaborative filtering treats ratings as a user–item utility matrix.
3. User-based CF compares users; item-based CF compares items.
4. SVD decomposes the matrix into user preferences, factor strengths, and item features.
5. Missing ratings are predicted by matching user vectors with item vectors.